

NMAP: NETWORK MAPPER

Breve introduzione ad nmap:

Su workstation utilizzo il comando “man nmap” per ottenere il manuale ufficiale di nmap.

Questo viene descritto come un tool open source per un “Network Mapper”, ovvero progettato per l’enumerazione e l’esplorazione di host su un network. E’ stato ideato per scannerizzare una larga scala network velocemente e in maniera efficace (lavora a basso livello per ‘craftare’ i pacchetti necessari per ottenere risposte adatte alla enumerazione host e porte) riportando quali host sono disponibili, quali servizi e su quali porte con nome e versione, quali Sistemi operativi e di che versione e infine che tipo di firewall o filtri sono implementati, oltre che accettare script/plugin per ottenere ulteriori informazioni sull’obiettivo.

DESCRIPTION

Nmap (“Network Mapper”) is an open source tool for network exploration and security auditing. It was designed to rapidly scan large networks, although it works fine against single hosts. Nmap uses raw IP packets in novel ways to determine what hosts are available on the network, what services (application name and version) those hosts are offering, what operating systems (and OS versions) they are running, what type of packet filters/firewalls are in use, and dozens of other characteristics. While Nmap is commonly used for security audits, many systems and network administrators find it useful for routine tasks such as network inventory, managing service upgrade schedules, and monitoring host or service uptime.

Ecco un esempio di comando per lo scan di nmap. Vediamo ora in dettaglio cosa fa esattamente.

Example 1. A representative Nmap scan

```
# nmap -A -T4 scanme.nmap.org

Nmap scan report for scanme.nmap.org (74.207.244.221)
Host is up (0.029s latency).
rDNS record for 74.207.244.221: li86-221.members.linode.com
Not shown: 995 closed ports
PORT      STATE      SERVICE      VERSION
22/tcp    open      ssh          OpenSSH 5.3p1 Debian 3ubuntu7 (protocol 2.0)
|_ ssh-hostkey: 1024 8d:60:f1:7c:ca:b7:3d:0a:d6:67:54:9d:69:d9:b9:dd (DSA)
|_ 2048 79:f8:09:ac:d4:e2:32:42:10:49:d3:bd:20:82:85:ec (RSA)
80/tcp    open      http         Apache httpd 2.2.14 ((Ubuntu))
|_ http-title: Go ahead and ScanMe!
646/tcp   filtered  ldap
1720/tcp  filtered  H.323/Q.931
9929/tcp  open      nping-echo   Nping echo
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6.39
OS details: Linux 2.6.39
Network Distance: 11 hops
Service Info: OS: Linux; CPE: cpe:/o:linux:kernel
```

Timing-related Options

Port scanning is often the most time consuming part of an Nmap scan (which might also include OS detection, version detection, and NSE scripts). While Nmap tries to be quick and efficient by default, manual optimization often helps. Nmap offers dozens of options for tailoring scan intensity and speed to match your exact needs. This section lists the most important options for optimizing port scan times. Options which take an amount of time are in seconds by default, or you may append *ms* (milliseconds), *s* (seconds), *m* (minutes), or *h* (hours) to the value. For further details on any of these options, see [the section called "Timing and Performance"](#). A much more thorough treatment, with examples and best-practices for improving Nmap performance is available in [Chapter 6, Optimizing Nmap Performance](#).

Top port scan performance options

-T0 through -T5

These timing templates affect many variables, offering a simple way to adjust overall Nmap speed from very slow (-T0) to extremely aggressive (-T5). A timing template may be combined with the more granular options describe below, and the most granular option takes precedence.

Il flag -T (Timing) indica la velocità con la quale i pacchetti raggiungono gli host per ottimizzare i tempi di velocità in corrispondenza del tipo di host e di percettibilità a scan. T0 (paranoid) è il più lento per maggiore evasione ma è estremamente lento, T5 è invece estremamente aggressivo e veloce ma altrettanto individuabile. -T4 è quindi un timing aggressivo, in questa applicazione per l'esercizio andrà bene così.

-A (Aggressive scan options)

This option enables additional advanced and aggressive options. Presently this enables OS detection (-o), version scanning (-sV), script scanning (-sC) and traceroute (--traceroute). More features may be added in the future. The point is to enable a comprehensive set of scan options without people having to remember a large set of flags. However, because script scanning with the default set is considered intrusive, you should not use -A against target networks without permission. This option only enables features, and not timing options (such as -T4) or verbosity options (-v) that you might want as well. Options which require privileges (e.g. root access) such as OS detection and traceroute will only be enabled if those privileges are available.

Il flag -A è l'opzione di scan aggressiva per quanto riguarda il TIPO di scan e NON il timing/frequenza/velocità. Nonostante anche questo abbia impatto su detection in maniera negativa, questo dipende dal tipo di scripts/richieste formulate anziché il numero stesso di pacchetti. E' l'equivalente di fare scan di OS (-o), di versione servizi (-sV), script default (-sC) e traceroute (--traceroute)

In questo caso non specifica range di porte (con flag -p) quindi verrà scannerizzato il default (top 1000).

Il target è scanme.nmap.org un sito configurato appositamente per permettere a giovani studenti di cybersecurity di imparare a usare le funzioni di base di nmap.



Nmap ScanMe

<http://scanme.nmap.org> · Traduci questa pagina



Go ahead and ScanMe! - Nmap

Hello, and welcome to **Scanme.Nmap.Org**, a service provided by the Nmap Security Scanner Project. We set up this machine to help folks learn about Nmap and ...

```
[analyst@secOps ~]$ nmap -A -T4 localhost
Starting Nmap 7.97 ( https://nmap.org ) at 2026-05-15 10:26 -0400
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000076s latency).
Other addresses for localhost (not scanned): ::1
Not shown: 998 closed tcp ports (conn-refused)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 2.0.8 or later
| ftp-syst:
|   STAT:
| FTP server status:
|   Connected to 127.0.0.1
|   Logged in as ftp
|   TYPE: ASCII
|   No session bandwidth limit
|   Session timeout in seconds is 300
|   Control connection is plain text
|   Data connections will be plain text
|   At session startup, client count was 2
|   vsFTPD 3.0.5 - secure, fast, stable
|_End of status
| ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_ -Iw-r--r-- 1 0 0 0 Mar 26 2018 ftp_test
22/tcp    open  ssh      OpenSSH 10.0 (protocol 2.0)
Service Info: Host: Welcome

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.30 seconds
```

Proviamo su localhost (noi stessi). Possiamo subito notare alcune informazioni chiave come la presenza del servizio ftp vsftpd (versione maggiore a 2.0.8, verrà riportata dopo come 3.0.5 nell'header) su porta 21/tcp.

Inoltre torna a noi l'informazione che la possibilità di login anonimo ftp è attivo con read-only permissions. Su porta 22/tcp troviamo invece servizio OpenSSH 10.0.

Quindi un servizio di File Transfer Protocol per condivisione file e Secure Shell protocol per accedere e configurare in remoto.

Proviamo a trovare il subnet di appartenenza della VM stessa e facciamo una scan della rete intera (in questo caso il range privato del NAT gestito da virtualbox su if: enp0s3)

```
[analyst@secOps ~]$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:2f:87:a7 brd ff:ff:ff:ff:ff:ff
    altname enx0800272f87a7
    inet 10.0.2.15/24 metric 1024 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 85564sec preferred_lft 85564sec
    inet6 fd17:625c:f037:2:a00:27ff:fe2f:87a7/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86350sec preferred_lft 14350sec
    inet6 fe80::a00:27ff:fe2f:87a7/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
```

```
[analyst@secOps ~]$ nmap -A -T4 10.0.2.0/24
Starting Nmap 7.97 ( https://nmap.org ) at 2026-05-15 10:34 -0400
Nmap scan report for 10.0.2.15
Host is up (0.00010s latency).
Not shown: 998 closed tcp ports (conn-refused)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 2.0.8 or later
| ftp-syst:
|   STAT:
| FTP server status:
|   Connected to 10.0.2.15
|   Logged in as ftp
|   TYPE: ASCII
|   No session bandwidth limit
|   Session timeout in seconds is 300
|   Control connection is plain text
|   Data connections will be plain text
|   At session startup, client count was 2
|   vsFTPD 3.0.5 - secure, fast, stable
|_End of status
| ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_rw-r--r--  1 0      0      0 Mar 26  2018 ftp_test
22/tcp    open  ssh      OpenSSH 10.0 (protocol 2.0)
Service Info: Host: Welcome

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 256 IP addresses (1 host up) scanned in 57.06 seconds
```

Unico host trovato e scannerizzato sembra essere quello della VM stessa 10.0.2.15.

Il gateway e DNS gestiti da virtualbox sono istruiti a non rispondere a scan (drop) perciò abbiamo identificato solo la nostra stessa macchina.

Potremmo trovare ulteriori dispositivi con host discovery (nmap -sn, ping sweep).

```
[analyst@secOps ~]$ nmap -A -T4 scanme.nmap.org
Starting Nmap 7.97 ( https://nmap.org ) at 2026-05-15 10:38 -0400
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.27s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 996 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   1024 ac:00:a0:1a:82:ff:cc:55:99:dc:67:2b:34:97:6b:75 (DSA)
|   2048 20:3d:2d:44:62:2a:b0:5a:9d:b5:b3:05:14:c2:a6:b2 (RSA)
|   256 96:02:bb:5e:57:54:1c:4e:45:2f:56:4c:4a:24:b2:57 (ECDSA)
|_  256 33:fa:91:0f:e0:e1:7b:1f:6d:05:a2:b0:f1:54:41:56 (ED25519)
53/tcp    open  domain   dnsmasq 2.84rc2
| dns-nsid:
|_  bind.version: dnsmasq-2.84rc2
80/tcp    open  http      Apache httpd 2.4.7 ((Ubuntu))
|_http-server-header: Apache/2.4.7 (Ubuntu)
|_http-title: Go ahead and ScanMe!
|_http-favicon: Nmap Project
9929/tcp  open  nping-echo Nping echo
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 33.85 seconds
```

Proviamo a scannerizzare scanme.nmap.org e vediamo invece i risultati.

IPv4 di scanme.nmap.org – 45.33.32.156

IPv6 di scanme.nmap.org - 2600:3c01::f03c:91ff:fe18:bb2f

Port aperti -

22/tcp open SSH OpenSSH 6.6.1p1

ssh-hostkeys per fingerprinting server ssh

53/tcp open DNS dnsmasq 2.84rc2

80/tcp open http Apache httpd 2.4.7

9929/tcp open nping-echo Nping echo #protocollo diagnostica

OS: Linux

```
[analyst@secOps ~]$ nmap -A -T4 scanme.nmap.org
Starting Nmap 7.97 ( https://nmap.org ) at 2026-05-15 10:38 -0400
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.27s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 996 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   1024 ac:00:a0:1a:82:ff:cc:55:99:dc:67:2b:34:97:6b:75 (DSA)
|   2048 20:3d:2d:44:62:2a:b0:5a:9d:b5:b3:05:14:c2:a6:b2 (RSA)
|   256 96:02:bb:5e:57:54:1c:4e:45:2f:56:4c:4a:24:b2:57 (ECDSA)
|_  256 33:fa:91:0f:e0:e1:7b:1f:6d:05:a2:b0:f1:54:41:56 (ED25519)
53/tcp    open  domain       dnsmasq 2.84rc2
| dns-nsid:
|_  bind.version: dnsmasq-2.84rc2
80/tcp    open  http         Apache httpd 2.4.7 ((Ubuntu))
|_http-server-header: Apache/2.4.7 (Ubuntu)
|_http-title: Go ahead and ScanMe!
|_http-favicon: Nmap Project
9929/tcp  open  nping-echo   Nping echo
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 33.85 seconds
```